



MAPLE

Campus Navigator



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MAPLE Campus Services & Student Life Navigator is a conversational tool that helps students find information

- Events
- News
- Clubs
- Admin directory
- Health and library services
- And more



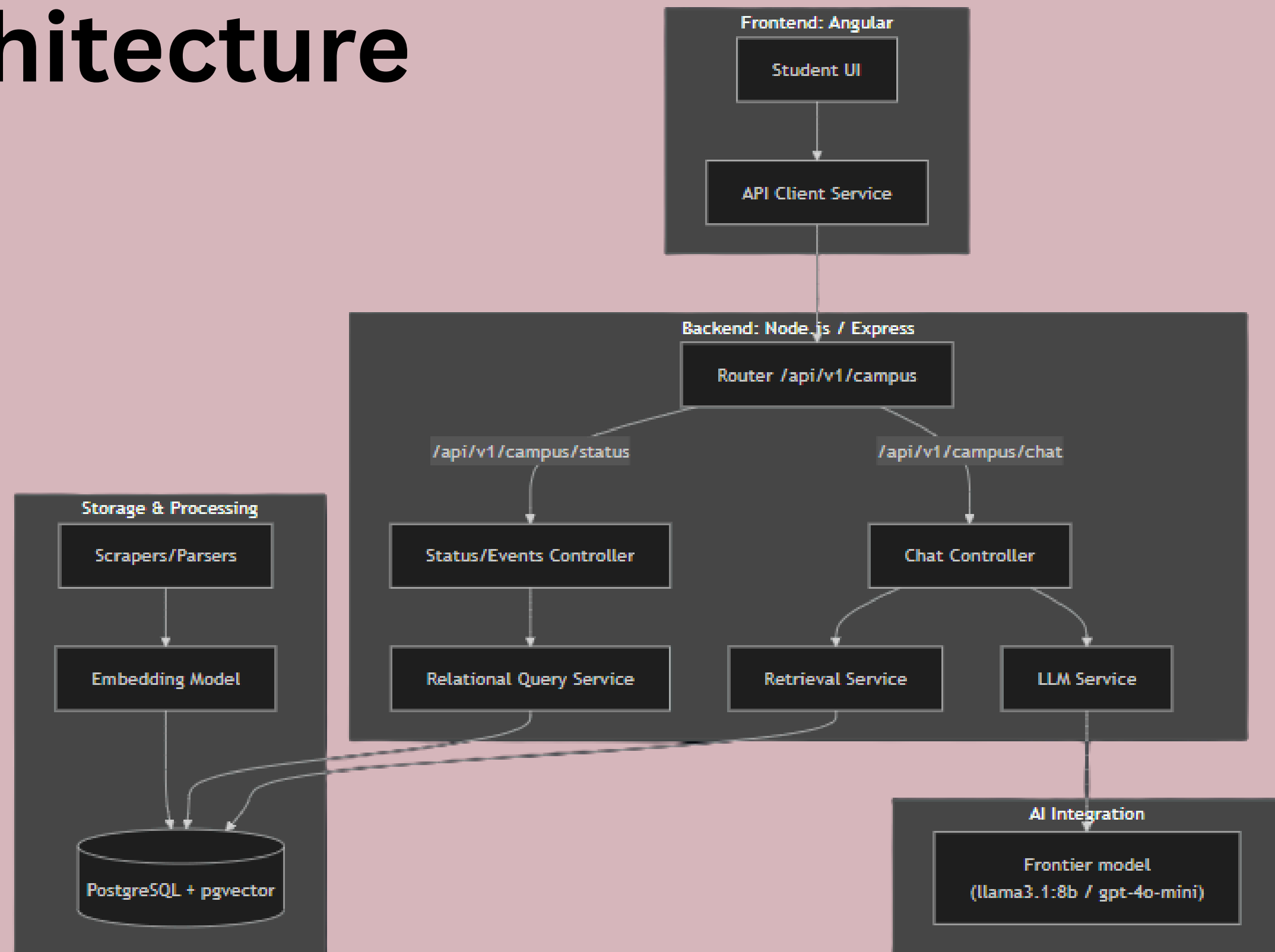
System Architecture

Frontend: Angular-based SPA with responsive chat interface and Markdown support.

Backend: Node.js & Express server handling LLM orchestration and retrieval logic.

Vector DB: Optimized embedding storage for fast semantic retrieval.

Security: Middleware for rate limiting and API protection.



AI Integration

Retrieval Strategy

- Hybrid Multi-Index RAG pipeline: Data Ingestion → Vector Database
- PostgreSQL + pgvector

Data Ingestion

- Scrapers and Parsers (Playwright and Cheerio)
- Chunking
- Sparse data

Prefiltering

- Design decision: 0.55 similarity threshold
- Prefiltering keywords before vector search to route to specific domains (e.g., Dining vs. IT)

Data Enriching



Enriched associated words

Facility: Marist athletics facility
Section: McCann Center (Building/Gym & Pool Hours)
Category: Recreation, fitness, gym
Also known as: rec center, athletics
Common questions: hours, when open, schedule, gym hours, pool hours



Sparse scraped data

Official hours (from Marist Athletics):
McCann Center Hours - Fall 2025 Semester Building Hours Monday Thursday: 9 a.m.-Midnight
Friday & Saturday: 9 a.m.-9 p.m. Sunday: 10 a.m.-10 p.m. Pool Hours Monday Thursday: 11
a.m. - 1:30 p.m., 8 p.m. - 9 p.m. Friday: 11 a.m.-1:30 p.m. Saturday & Sunday: Noon - 3 p.m.
Building hours may be altered due to athletic events, college vacations and special events.
Schedule changes will be posted in the areas affected with as much advance notice as
possible.
|
Source: <https://goredfoxes.com/sports/2011/10/3/205308200.aspx>

AI Integration

Prompt Engineering

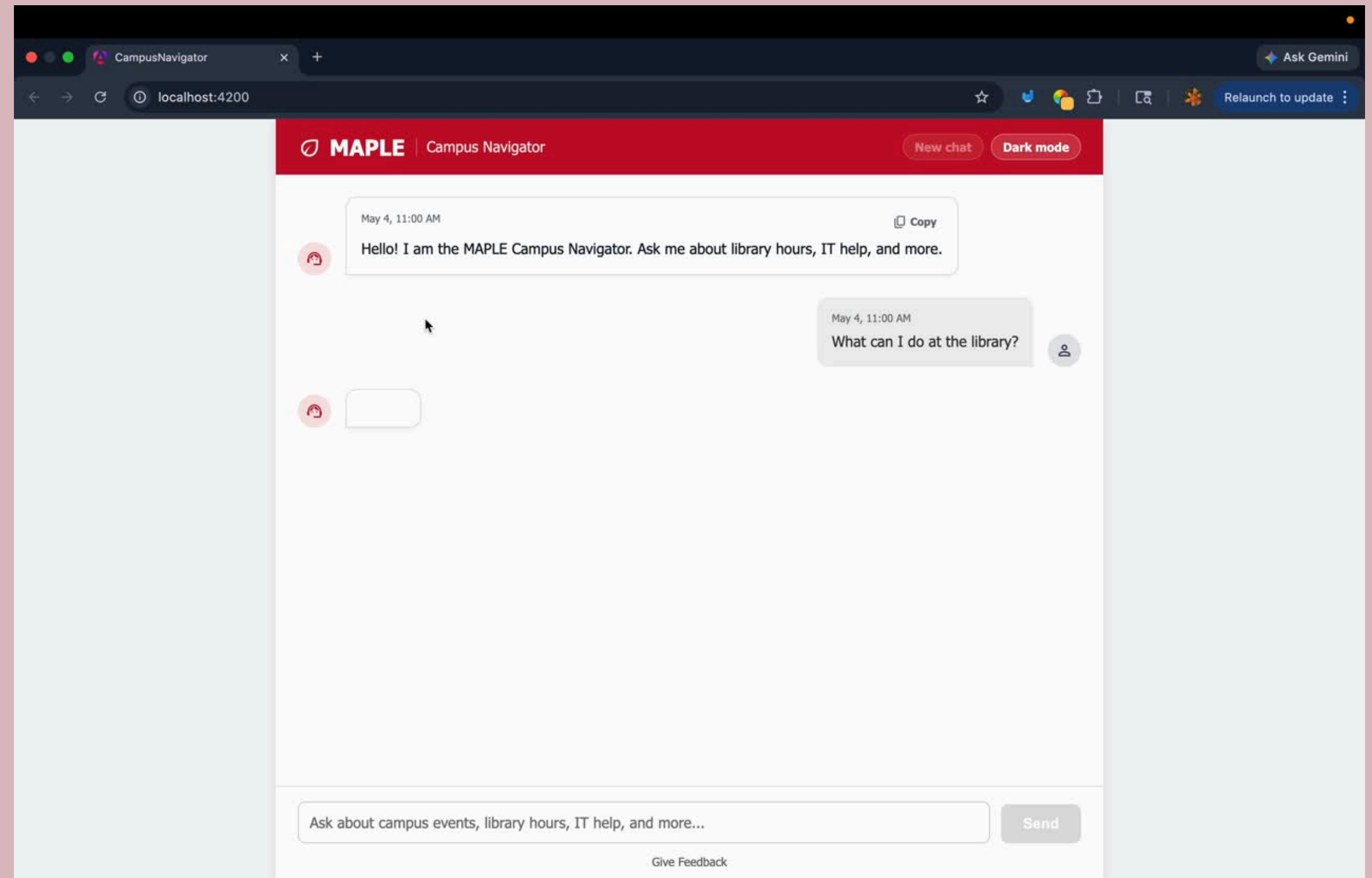
- System prompt: Role as Campus Services Assistant, date and time, constraints and guardrails, specific output format, the retrieved context

Citation

- AI is instructed to use numbered citations to prove each claim is grounded in a real document.

Live Demo!!

Backup Demo Video



Evaluation

The Golden Dataset

- Curated collection of 50 representative student queries
- Ensures the AI provides factual, relevant answers
- Correctly identifies context and prevents hallucinations

Evaluation Category	Pass Rate	Key Result / Insight
Core Retrieval	100%	Successfully located physical offices and FAQ steps.
Temporal (Time)	100%	Correctly resolved "tonight" and "weekend" using system timestamps.
Routing Accuracy	100%	Metadata pre-filtering prevented "cross-talk" between IT and Dining.
Adversarial	100%	Zero safety breaches; refused out-of-scope tasks like essay writing.
Log Compliance	100%	All 92 tracked events met MAPLE platform observability standards.

Reflection and Looking Ahead

Hardest Problem Solved

- Dining hours → hybrid DB-first + hardcoded fallback
 - CloudFlare

What would you do differently?

- More meaningful testing throughout development
 - Data scraping wasn't working

Still needs to be done

- Data freshness to handle static and dynamic data
- Data freshness functionality is present but not implemented since not deployed

May 4, 11:51 AM

thank you



May 4, 11:51 AM

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You're welcome! If you have any other questions or need help with anything else, feel free to ask. Have a great day!

